

Microsoft Fabric

A Practical Basics Guide for Analysts working with Power BI & Excel

Concepts • Architecture • Diagrams • Examples • Interview Q&A

1. What is Microsoft Fabric?

Microsoft Fabric is an all-in-one, cloud-based (SaaS) analytics platform from Microsoft that brings together data engineering, data integration, data warehousing, data science, real-time analytics, and business intelligence (Power BI) into a **single product with one underlying store of data**. Instead of provisioning and stitching together separate Azure services (Data Factory, Synapse, Databricks-like compute, Power BI) yourself, Fabric gives you all of these as built-in "experiences" inside one workspace, all reading and writing the same data.

Why this matters to you as a Power BI / Excel analyst: Today you probably import or connect to data that someone else has already extracted, cleaned, and loaded somewhere. In Fabric, the same lake that data engineers use to store tables is the one your Power BI report and even your Excel workbook can connect to directly — often with **zero data duplication** (via a mode called Direct Lake, explained in Section 8).

1.1 The problem Fabric solves

Before Fabric, a typical analytics stack needed: one tool to move data (ETL), one to store it (a data lake or warehouse), one for data science, one for real-time streaming, and Power BI for reporting — each with its own billing, security model, and copy of the data. This caused: multiple copies of data (which drift out of sync), complex security to maintain in many places, and separate bills/capacity to manage. Fabric's core idea is: **one copy of the data, one security model, one capacity, many tools on top**.

1.2 Key characteristics

- **SaaS (Software as a Service):** No infrastructure to install, patch, or scale manually — you just turn it on for your Microsoft 365 / Power BI tenant.
- **Unified storage (OneLake):** Every workload stores its data in one lake, in an open format (Delta Parquet).
- **Unified experience:** Data engineers, data scientists, and BI analysts all work inside the same portal and workspace concept used by Power BI today.
- **One licensing model:** A single "Fabric capacity" (F SKU) powers every workload, instead of separate bills per service.

2. Fabric Architecture at a Glance

Microsoft Fabric is best understood as a stack: **OneLake** is the single storage foundation at the bottom, and on top of it sit several purpose-built "experiences" (workloads), each aimed at a different persona (engineer, scientist, analyst). Power BI — the tool you already know — is simply one of these experiences, now reading data straight from the same lake.

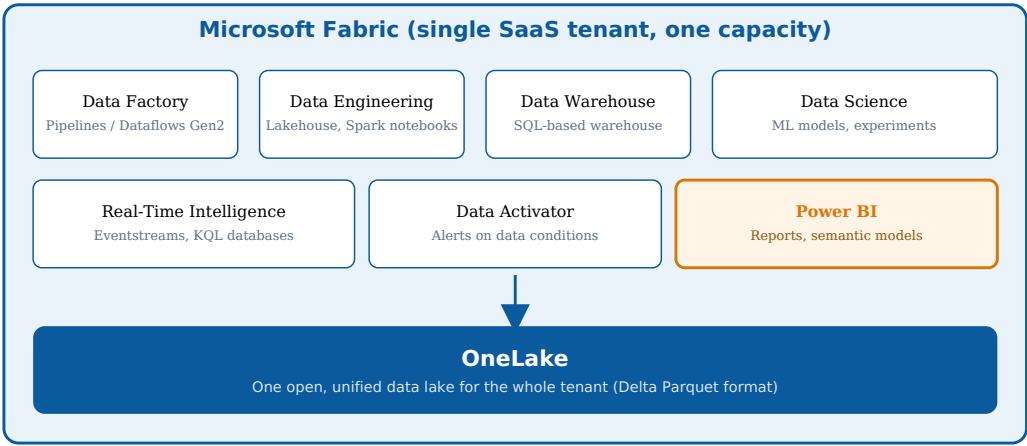


Fig. 1 — All Fabric workloads, including Power BI, sit on top of and read/write the same OneLake storage.

2.1 The persona → workload map

Persona	Fabric Workload	What they do with it
Data Engineer	Data Engineering (Lakehouse) & Data Factory	Build pipelines, clean and transform raw data into governed Delta tables
Data Analyst / Report Analyst (you)	Power BI, Data Warehouse, Dataflows Gen2	Build semantic models, DAX measures, reports and dashboards; light transforms with Power Query
Data Scientist	Data Science	Train and track ML models on the same lake data using notebooks
App/Ops Developer	Real-Time Intelligence, Data Activator	Stream events (IoT, logs) and trigger automated alerts/actions

3. OneLake — the Foundation

Think of **OneLake** as "**OneDrive for data.**" Just as every user in an organisation gets one OneDrive and every Microsoft 365 file ultimately lives in SharePoint/OneDrive, every Fabric tenant automatically gets exactly one OneLake. Every workspace you create gets a folder inside it, and every Lakehouse/Warehouse/dataset you build stores its actual data files there — in an open, non-proprietary format called **Delta Parquet** (the same format used by Databricks and open-source Delta Lake).

3.1 Key properties of OneLake

- **One per tenant:** Automatically provisioned — nothing to set up.
- **Open format:** Data is stored as Delta tables (Parquet files + a transaction log), so non-Microsoft tools (Spark, Python, even direct file access) can read it too.
- **Shortcuts:** Instead of copying data into OneLake, you can create a "shortcut" — a pointer to data that physically still lives in another Lakehouse, an Azure Data Lake Storage (ADLS) Gen2 account, or even Amazon S3/Google Cloud Storage. It appears as if the data is local, with no duplication or extra ETL job needed.
- **One security model:** Access control (OneLake data access roles) is applied once, and every engine (Spark, SQL, Power BI) that touches that data honours it.

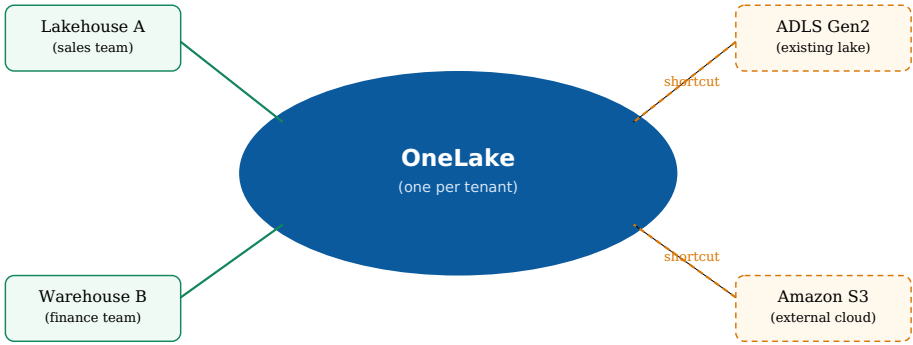


Fig. 2 — Solid lines: data physically stored in OneLake. Dashed lines: shortcuts pointing to data that stays where it is, avoiding duplication.

Analogy recap: OneDrive : files :: OneLake : data tables. Workspaces are like folders; shortcuts are like "add shortcut to OneDrive" for a SharePoint folder you don't own.

4. Workspaces & Capacities

If you already use Power BI, **workspaces** will feel familiar — they are still the container you publish reports into and share with colleagues. In Fabric, a workspace can now also hold Lakehouses, Warehouses, notebooks, pipelines, and ML models, not just reports and datasets.

4.1 Capacity (the "engine size")

A **capacity** is the pool of compute power (CPU/memory, measured in "Capacity Units" or CUs) that all the items in a workspace share when they run queries, refresh data, or execute pipelines. Every workspace is assigned to exactly one capacity. Fabric capacities are named with an **F-SKU** (F2, F4, F8 ... up to F2048); a higher number means more power and a higher price. Power BI Premium capacities (P SKUs) and Premium-Per-User are being aligned with these F SKUs.

SKU	Capacity Units (CU)	Typical fit
F2 / F4	2 / 4	Personal trial, small team, learning
F64	64	Roughly equivalent to a Power BI Premium P1 — common starting point for mid-size orgs
F128 - F512	128 - 512	Larger organisations, heavier ETL/Spark workloads
F1024 / F2048	1024 / 2048	Enterprise-scale, many concurrent heavy workloads

Fabric trial: Microsoft offers a 60-day free Fabric trial capacity so you can try every workload (Lakehouse, Warehouse, Data Science, etc.) without a paid subscription — the easiest way to start learning hands-on.

5. Core Fabric Workloads Explained

5.1 Data Factory (pipelines & Dataflows Gen2)

The data-movement and transformation layer. **Pipelines** orchestrate "copy data from A to B, then run this notebook, then refresh that dataset" style workflows (similar to Azure Data Factory). **Dataflows Gen2** are the evolution of Power Query — if you already build queries in Power BI Desktop or Excel's Power Query editor, Dataflows Gen2 use the exact same visual, no-code/low-code interface, but the output lands as a table in a Lakehouse/Warehouse that anyone can reuse, instead of being locked inside one report.

5.2 Data Engineering — the Lakehouse

A **Lakehouse** combines the flexibility of a data lake (store any file: CSV, JSON, images, plus structured tables) with the manageability of a database (SQL queries, a schema, relationships). Under the hood, its tables are Delta Parquet files stored in OneLake. Data engineers use **Spark notebooks** (Python/PySpark, Scala, or Spark SQL) to clean and reshape data here.

5.3 Data Warehouse

A fully relational, T-SQL-based data warehouse — for teams who prefer writing standard SQL (like SQL Server / Synapse SQL) over Spark notebooks. Both a Lakehouse and a Warehouse store data as Delta Parquet in OneLake, so a report built on either behaves the same way to Power BI.

5.4 Data Science

Notebooks and experiment tracking (MLflow-based) for building, training and scoring machine-learning models directly against Lakehouse data — no separate export/import step needed.

5.5 Real-Time Intelligence

Eventstreams capture continuously arriving data (IoT sensors, clickstreams, application logs) and route it into a **KQL Database** (Kusto Query Language — the engine behind Azure Data Explorer), optimised for fast queries over huge volumes of time-stamped events.

5.6 Data Activator

A no-code tool that watches data (e.g., a Power BI report's numbers, or a real-time stream) and automatically fires an alert or action when a condition is met — e.g., "notify the regional manager whenever daily sales drop more than 15% versus last week."

5.7 Power BI in Fabric

Exactly the Power BI you know (reports, dashboards, semantic models/datasets), now able to connect directly to Lakehouse/Warehouse tables in OneLake with no import step, via **Direct Lake mode** (Section 8).

6. Lakehouse vs. Data Warehouse — Which One?

Aspect	Lakehouse	Data Warehouse
Primary language	Spark notebooks (PySpark/Scala/Spark SQL); also supports a SQL analytics endpoint	T-SQL only
Best for	Data engineers, unstructured + structured data, big-data transforms	Analysts/BI teams comfortable with SQL, strict relational modelling
File types supported	Any file (images, CSV, JSON, Parquet) plus managed Delta tables	Structured tables only
Underlying storage	Delta Parquet in OneLake	Delta Parquet in OneLake (same!)
Transactions (ACID)	Yes, via Delta Lake	Yes, full T-SQL transactional guarantees

Good to know: Because both store data the same way in OneLake, it's common to see hybrid architectures — engineers land and clean data in a Lakehouse, while analysts query it through a Warehouse's SQL endpoint, with no data actually copied between the two.

7. Delta Lake Format — Why it Matters

Every table in OneLake is stored as a **Delta table**: a folder of Parquet files (compressed, columnar — very fast for analytics) plus a **transaction log** that records every insert/update/delete, giving you:

- **ACID transactions** — reliable updates, no half-written data even if a job fails midway.
- **Time travel** — query a table "as of" an earlier version.
- **Schema enforcement/evolution** — controlled way to add new columns over time.
- **Open access** — any engine that understands Delta (Spark, Power BI, even Python `pandas` via `delta-rs`) can read it without conversion.

This open format is precisely what makes Direct Lake mode possible for Power BI — the engine reads the Parquet files straight off OneLake instead of needing its own private copy.

8. Power BI Connectivity Modes — Import vs. DirectQuery vs. Direct Lake

This is the single most important concept for a Power BI analyst moving into Fabric. Your semantic model (dataset) can get its data in three fundamentally different ways:

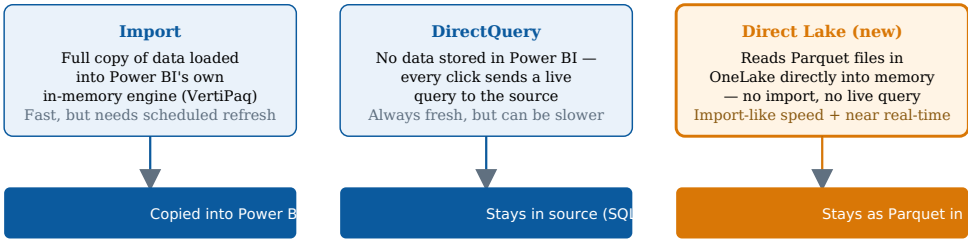


Fig. 3 — Direct Lake gives Import-mode-like performance without a refresh copy, by reading OneLake's Parquet files straight into memory on demand.

Mode	Data location	Freshness	Typical use
Import	Copied into the dataset's own in-memory store	As of last scheduled refresh	Most traditional Power BI reports today
DirectQuery	Stays in the source system	Real-time (query runs live)	Huge/fast-changing sources where a full copy is impractical
Direct Lake	Stays as Delta Parquet in OneLake, read on demand	Near real-time (as soon as new Parquet files land)	Fabric Lakehouse/Warehouse-backed reports at large scale, avoiding both refresh lag and DirectQuery slowness

Note: Direct Lake can silently "fall back" to DirectQuery for certain unsupported query patterns (e.g., some complex DAX). Good model design (star schema, appropriate data types) keeps it in true Direct Lake mode for best performance.

9. Dataflows Gen2 — Power Query, Levelled Up

If you build queries in Power BI Desktop or Excel's Power Query editor today, Dataflows Gen2 will feel almost identical: the same ribbon, the same M language under the hood, the same step-by-step applied transformations. The difference in Fabric is the **destination**: instead of the transformed data being locked inside one Power BI file, you explicitly pick a destination (a Lakehouse or Warehouse table), so the cleaned data becomes a reusable, governed asset that any report, notebook, or even another dataflow can build on.

10. Excel + Fabric — What Changes for You

Excel connects to Fabric data in a few practical ways:

- **Analyze in Excel / "Explore in Excel"**: From a Power BI report or a Fabric item, you can open a live, PivotTable-style connection in Excel that queries the underlying semantic model directly — no copy-paste, no static export.
- **SQL analytics endpoint**: Every Lakehouse and Warehouse automatically exposes a SQL endpoint. In Excel, you can use `Get Data → From Database → SQL Server` and point it at that endpoint to pull Fabric tables straight into Excel, just like connecting to any SQL Server today.
- **Power Query in Excel**: The same Power Query engine used in Dataflows Gen2 is built into Excel, so M-language knowledge transfers directly between the two.
- **OneLake file explorer / Excel add-in**: Lets you browse OneLake like a file system from within Excel/Windows Explorer.

10.1 End-to-End Example

A concrete, common scenario tying everything together:

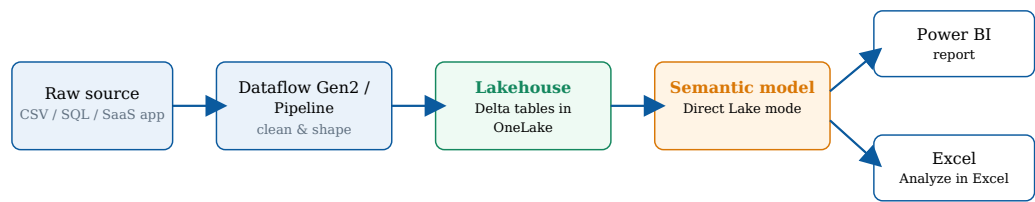


Fig. 4 — A sales CSV lands in a Lakehouse via a Dataflow, becomes a Direct Lake semantic model, and is consumed by both a Power BI report and an Excel PivotTable, with only one physical copy of the data.

1. A regional sales CSV is uploaded and shaped using a **Dataflow Gen2** (same Power Query skills you already have).
2. The clean, structured output lands as a Delta table inside a **Lakehouse** — physically stored once, in OneLake.
3. A **semantic model** is built on top in **Direct Lake** mode — no import refresh needed, DAX measures added as usual.
4. A **Power BI report** is published from that model for the wider team.
5. You, the analyst, use **Analyze in Excel** to build a PivotTable against the exact same model for ad-hoc analysis — always in sync with the report, no separate extract to maintain.

11. Security & Governance Basics

Layer	What it controls
Workspace roles	Admin / Member / Contributor / Viewer — control who can edit vs. just view items in a workspace (same roles as Power BI today)
OneLake data access roles	Fine-grained, table/folder-level read access inside a Lakehouse, enforced consistently for every engine that reads it
Sensitivity labels	Microsoft Purview labels (e.g., "Confidential") can be applied to Fabric items and flow down to exports, e.g. into Excel
Tenant settings	Org-wide switches (set by a Fabric/Power BI admin) that turn features like external sharing or specific workloads on/off

12. Getting Started — First Steps

1. Sign in at `app.fabric.microsoft.com` (or the Fabric switch inside `app.powerbi.com`) and start (or ask an admin to start) the 60-day **free trial capacity**.
2. Create a **Workspace** and assign it to your trial/F-SKU capacity.
3. Create a **Lakehouse**, upload a sample CSV, and load it as a Delta table.
4. Open the auto-generated **SQL analytics endpoint** and run a simple `SELECT` query to see the same data available via SQL.
5. Build a **Direct Lake semantic model** on top and create a quick Power BI report.
6. Try **Analyze in Excel** from that report to see the live PivotTable connection.

13. Quick Glossary

Term	Meaning
OneLake	The single, tenant-wide data lake underlying all of Fabric
Lakehouse	A Fabric item combining lake-style file storage with database-style Delta tables, queried via Spark or SQL
Warehouse	A fully T-SQL relational warehouse item, also stored as Delta Parquet in OneLake
Shortcut	A pointer inside OneLake to data stored elsewhere (another Fabric item, ADLS Gen2, S3, etc.) — no data copy
Delta table	Parquet files + transaction log giving ACID transactions, schema control and time travel
Direct Lake	Power BI storage mode that reads OneLake Parquet files directly into memory, skipping import refresh
Capacity (F-SKU)	The purchased compute pool (e.g., F64) that powers every workload in the workspaces assigned to it
Dataflow Gen2	Power Query-based, no-code data prep tool that writes its output into a Lakehouse/Warehouse table
Semantic model	The Fabric/Power BI term for what used to be called a "dataset" — tables, relationships and DAX measures
KQL Database	A database optimised for querying huge volumes of time-stamped/event data using Kusto Query Language

14. Top 10 Interview Questions on Microsoft Fabric

1 What is Microsoft Fabric, and how is it different from using separate Azure services like Data Factory, Synapse, and Power BI individually?

Fabric is a single SaaS analytics platform that unifies data engineering, warehousing, data science, real-time analytics and BI under one roof, all backed by one shared lake (OneLake) and one capacity/billing model. The key difference from stitching together separate Azure services is that Fabric removes duplicate data copies, unifies security, and lets every workload share the same governed data instead of each service holding its own.

2 What is OneLake, and why is it often compared to OneDrive?

OneLake is the single, automatically provisioned data lake that underlies the entire Fabric tenant — every workspace and item stores its data there in open Delta Parquet format. It's compared to OneDrive because, just like every user gets one OneDrive without provisioning storage manually, every tenant gets exactly one OneLake, and "shortcuts" let you reference data elsewhere without duplicating it, similar to adding a shortcut to a shared folder in OneDrive.

3 Explain the difference between Import, DirectQuery and Direct Lake modes in Power BI.

Import copies all data into Power BI's own in-memory engine, giving fast performance but requiring scheduled refreshes. DirectQuery leaves data in the source and sends a live query for every interaction, keeping data fresh but potentially slower. Direct Lake, available only against OneLake-backed tables, reads the Delta Parquet files directly into memory on demand — combining Import-like speed with near real-time freshness, without a refresh job.

4 What is the difference between a Lakehouse and a Data Warehouse in Fabric?

Both store their data as Delta Parquet tables in OneLake, so they're interoperable. A Lakehouse is aimed at data engineers: it accepts any file type and is queried mainly through Spark notebooks (plus a SQL endpoint). A Warehouse is a fully relational, T-SQL-only engine aimed at analysts who prefer standard SQL and stricter relational modelling.

5 What is a Fabric capacity, and what do the F-SKUs (F2, F64, F128, etc.) represent?

A capacity is the shared pool of compute power, measured in Capacity Units, assigned to one or more workspaces; every item inside those workspaces draws on it for refreshes, queries, and pipeline runs. F-SKUs (F2 up to F2048) represent increasing amounts of that compute power — a higher number means more concurrent capacity and a higher price, and they are designed to align with the older Power BI Premium P-SKU tiers.

6 What are Dataflows Gen2, and how do they relate to Power Query in Excel or Power BI Desktop?

Dataflows Gen2 are Fabric's evolution of Power Query: they use the same M-language engine and visual query editor you already use in Excel or Power BI Desktop, but instead of the transformed data staying inside one file, you choose a destination — typically a Lakehouse or Warehouse table — so the cleaned data becomes a reusable asset for any report, notebook, or other dataflow to build on.

7 What is a "shortcut" in OneLake, and why would you use one instead of copying data?

A shortcut is a reference inside OneLake that points to data physically stored elsewhere — another Fabric item, an Azure Data Lake Storage Gen2 account, or even Amazon S3 — so it appears local without duplicating storage or requiring an extra ETL job. It's used to avoid redundant copies, keep a single source of truth, and let multiple teams reuse the same data across workspaces without maintaining separate pipelines.

8 How would you connect Excel to data stored in a Fabric Lakehouse?

Two common ways: use "Analyze in Excel" (or "Explore in Excel") from an existing Power BI report/semantic model to get a live PivotTable connection, or connect Excel directly to the Lakehouse's auto-generated SQL analytics endpoint via Get Data → From Database → SQL Server, pulling the underlying Delta tables just like any SQL Server source.

9 What is Delta Lake/Delta Parquet format, and why is it central to Fabric?

Delta format stores data as Parquet files plus a transaction log, which adds ACID transactions, schema enforcement/evolution, and time travel on top of plain Parquet's fast, columnar storage. It's central to Fabric because every workload — Lakehouse, Warehouse, and Power BI's Direct Lake mode — reads and writes this same open format in OneLake, which is what allows different engines to share one physical copy of data reliably.

10 What roles and security controls are available to govern access to data in Fabric?

At the workspace level, standard roles (Admin, Member, Contributor, Viewer) control who can edit or view items, carried over from Power BI. Inside a Lakehouse, OneLake data access roles give finer-grained, table/folder-level permissions enforced consistently across every engine that touches the data. Sensitivity labels (via Microsoft Purview) can also be attached to items and follow the data into exports such as Excel, and tenant-level admin settings control which features (e.g., external sharing) are available org-wide.